Identity & Family Information

Sighting & Observation Information

Association & Analysis Information

POD

pod_id	**PK**
pod_name	
description	

MATRILINE

matriline_id	**PK**
pod_id	FK
matriline_name	
description	

WHALE

whale_id	**PK**
name	
sex	
birth_date	
death_date	
matriline_id	FK
notes	

KINSHIP

kinship_id	**PK**
subject_whale_id	FK
related_whale_id	FK
relationship_type (e.g motherOf, siblingOf)	
certainty (e.g confirmed, probable)	
source_id	FK
notes	

SIGHTING

sighting_id	**PK**
location_id	FK
source_id	FK
datetime	
notes	

LOCATION

location_id	**PK**
latitude	
longitude	
utm_x	
utm_y	
coordinate_system (e.g. WGS84, UTM)	
area (e.g Strait, Region)	
notes	

INDIVIDUAL_OBSERVATION

observation_id	**PK**
sighting_id	FK
whale_id	FK
source_id	FK
identification_status (e.g., confirmed, inferred)	
identification_basis (e.g., photo ID, group report)	
notes	

GROUP_OBSERVATION

group_obs_id	**PK**
sighting_id	FK
source_id	FK
reported_group (e.g., J, K, L, JKL, SRKW)	
group_type (e.g., pod, matriline, general)	
source detail	
notes	

DATA_SOURCE

source_id	**PK**
source_name (e.g CWR, Dryad, Whale Museum)	
source_type (e.g ID guide, dataset, sighting report)	
citation description, url	
timestamp_shifted	
coordinates_shifted	

Association

association_id	**PK**
whale_a_id	FK
whale_b_id	FK
association_type (e.g, co-sighting, predicted)	
score	
method (e.g., network analysis)	
time_window	
source_id	FK
notes	

contains (POD 1 — N MATRILINE)
includes (MATRILINE 1 — N WHALE)
has (POD 1 — N WHALE)
observed in (WHALE 0..1 — 0..N INDIVIDUAL_OBSERVATION)
subject whale (WHALE 1 — N KINSHIP)
related whale (WHALE 1 — N KINSHIP)
supported by (KINSHIP 0..N — 1 DATA_SOURCE)
occurs at (SIGHTING N — 1 LOCATION)
has (SIGHTING 1 — 0..N INDIVIDUAL_OBSERVATION)
has (SIGHTING 1 — 0..N GROUP_OBSERVATION)
recorded by (INDIVIDUAL_OBSERVATION N — 1 DATA_SOURCE)
recorded by (GROUP_OBSERVATION N — 1 DATA_SOURCE)
supported by (SIGHTING 0..N — 1 Association)
whale A (WHALE 1 — 0..N Association)
whale B (WHALE 1 — 0..N Association)
derived using (Association 0..N — 1 DATA_SOURCE)

Legend

- Identity & family entities
- Relationship entities
- Sighting & observation entities
- Provenance entites
- Association & analysis entities
- Relationship (diamond)

Notation

ENTITY	
attribute_name	**PK**
attribute_name	FK
attribute_name	
attribute_name	

PK = Primary Key
FK = Foreign Key

1 = one
N = many
0..1 = zero or one
0..N = zero or many

Key Modeling Decisions

1. Separate group reports from individual observations: A sighting can exist without known individual whales, Group observations store reported pod or group information,
2. Keep family relationships separate from recorded associations: Kinship (e.g. mother-offspring) represents biological relationships, while Association represents recorded co-sightings or derived association candidates.
3. Preserve sources and identification information: sightings, observations, kinship records, and association records link to their data sources. Individual observations retain identification status and basis. Source metadata records whether timestamps and coordinates were shifted.
4. Distinguish recorded evidence from potential association candidates: Association type distinguishes recorded co-sighting from potential candidates. Method, score, and time window describe derived results. Supporting sightings provide observation evidence and co-sightings alone do not establish close social relationships.
Note: An individual observation may have no identified whale. For motherOf, the subject whale is the mother and the related whale offspring.